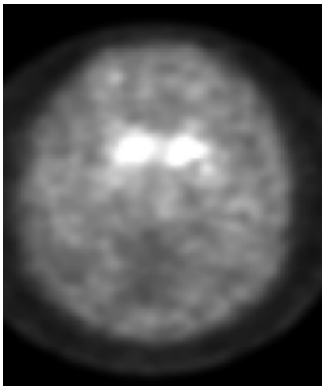
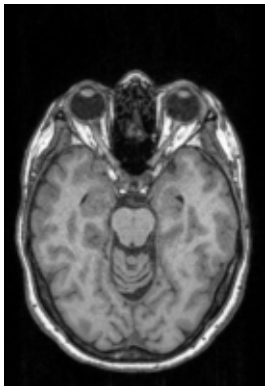


Subject: tutorial_subject

DaTSCAN



T1w



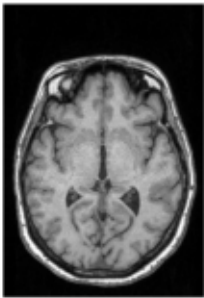
DaTSCAN	datscan.nii.gz
MRI / T1w	T1w.nii.gz
Report date	22-Jun-2026
Quality control	A

DaTSCAN / MRI qualitative images

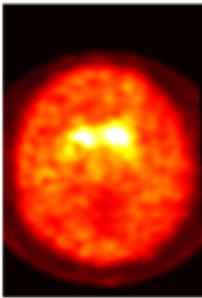
Axial view

Axial view

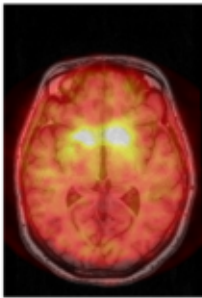
MRI / T1w



Registered DaTSCAN



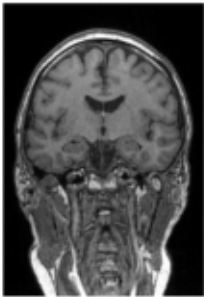
Overlay



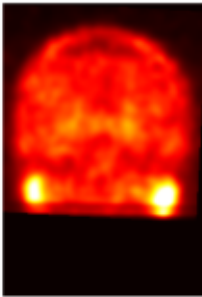
Coronal view

Coronal view

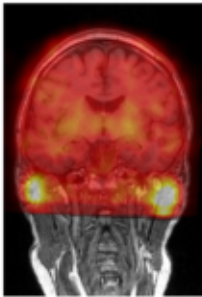
MRI / T1w



Registered DaTSCAN



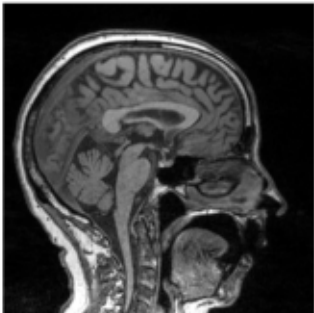
Overlay



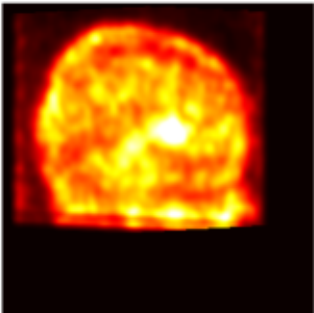
Sagittal view

Sagittal view

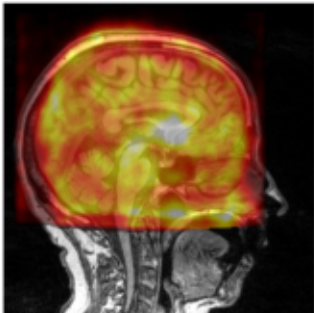
MRI / T1w



Registered DaTSCAN



Overlay



DaTSCAN quantitative biomarkers

Striatum nucleus uptake statistics from pipeline output (*_sbr.tsv)

Region	Right	Left	Bilateral
Striatum	12688.1357	13082.4131	12887.3809
Caudate	14056.3105	13238.3662	13658.9062
Putamen	11687.3896	12982.3848	12359.0879
Occipital reference	5734.8774	5633.6621	5685.0737

Note: values in this table are read from the backend TSV output. The Gradio app does not recompute additional biomarkers.

Classification

Healthy control / Parkinson's disease classification:

Not implemented in the current backend version.

MRI atrophy quantification

Atrophy quantification of MRI:

Not implemented in the current backend version.

Quality control

Quality control: A

A = good execution with generated biomarkers; B = execution completed but biomarkers require review; C = failed execution.

Complete SBR TSV output

id	processing	hemi	aggregate	loss	dat_str	dat_cau	dat_put	dat_occ
0	raw	R	sum	0.3191	108635816.0000	50841676.0000	57794144.0000	184651584.0000
0	raw	R	mean	0.3191	12688.1357	14056.3105	11687.3896	5734.8774
0	raw	R	median	0.3191	11615.4434	13519.2285	10724.0039	5701.0996
0	raw	R	std	0.3191	4071.3074	4891.4756	2971.1956	1213.3490
0	raw	R	p10	0.3191	8738.8066	7564.4009	8928.5801	4232.8374
0	raw	R	p30	0.3191	9854.2363	10993.1758	9562.1416	5186.2344
0	raw	R	p70	0.3191	14094.2354	16601.5605	12472.7695	6307.8101
0	raw	L	sum	0.3191	114431864.0000	45248736.0000	69183128.0000	175719552.0000
0	raw	L	mean	0.3191	13082.4131	13238.3662	12982.3848	5633.6621
0	raw	L	median	0.3191	12344.7148	13029.2227	11910.0273	5718.4062
0	raw	L	std	0.3191	3753.7114	3728.7415	3766.2422	1171.9257
0	raw	L	p10	0.3191	8438.9941	7755.5215	8760.3340	3900.6289
0	raw	L	p30	0.3191	10660.7910	11143.5928	10457.7832	5172.8657
0	raw	L	p70	0.3191	15059.8418	15365.5820	14718.4189	6322.0049
0	raw	Both	sum	0.3191	223067680.0000	96090408.0000	126977264.0000	360371136.0000
0	raw	Both	mean	0.3191	12887.3809	13658.9062	12359.0879	5685.0737
0	raw	Both	median	0.3191	11992.8564	13228.9316	11350.4160	5709.7056
0	raw	Both	std	0.3191	3918.9954	4384.5078	3467.7173	1194.2186

Complete SBR TSV output

id	processing	hemi	aggregate	loss	dat_str	dat_cau	dat_put	dat_occ
0	raw	Both	p10	0.3191	8624.7959	7682.7354	8880.7598	4054.0120
0	raw	Both	p30	0.3191	10254.9199	11075.1816	9930.7881	5179.9971
0	raw	Both	p70	0.3191	14586.7061	15892.0615	13433.2227	6314.1470

Complete symmetry TSV output

id	metric	hemi	aggregate	dat_cau	dat_put
0	lncc	R	sum	20.4009	0.0000
0	lncc	R	std	0.0342	0.0000
0	lncc	R	mean	0.0056	0.0000
0	lncc	R	min	0.0000	0.0000
0	lncc	R	max	0.4890	0.0000
0	lncc	L	sum	1115.4106	954.5286
0	lncc	L	std	0.2345	0.1908
0	lncc	L	mean	0.3263	0.1791
0	lncc	L	min	0.0000	0.0000
0	lncc	L	max	0.8722	0.8125
0	lncc	Both	sum	1135.8116	954.5286
0	lncc	Both	std	0.2302	0.1640
0	lncc	Both	mean	0.1615	0.0929
0	lncc	Both	min	0.0000	0.0000
0	lncc	Both	max	0.8722	0.8125
0	l2	R	sum	9535.2461	10839.1641
0	l2	R	std	0.9174	0.5572
0	l2	R	mean	2.6362	2.1919
0	l2	R	min	0.9824	1.5068
0	l2	R	max	4.7135	4.3679

Complete symmetry TSV output

id	metric	hemi	aggregate	dat_cau	dat_put
0	I2	L	sum	4937.7275	5998.9463
0	I2	L	std	0.7985	0.7341
0	I2	L	mean	1.4446	1.1257
0	I2	L	min	0.0018	0.0001
0	I2	L	max	3.1549	3.1390
0	I2	Both	sum	14472.9727	16838.1113
0	I2	Both	std	1.0475	0.8443
0	I2	Both	mean	2.0573	1.6389
0	I2	Both	min	0.0018	0.0001
0	I2	Both	max	4.7135	4.3679

Execution information

Pipeline completed successfully.

Output folder:
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\gradio_runs\run_20260622_153043_tutorial_subject\output

SBR rows found: 21
Symmetry rows found: 30

Terminal output:

-----#
Running Dat2MRI algorithm #
-----#

Total sessions to process: N=1

Subject: 0 (0/1)
* Preprocessing.
* DaT symmetry and mask.
* MRI DaT simulation and mask.
* MRI-DaT registration.
Optimizer: LBFGS lr=0.1 max iter=10
Losses: reg(w=1); reg_label(w=2); reg_uptake(w=0.0); reg_lr(w=0.5); regularizer(w=1)

Training started #
#####

* Epoch: 0
Epoch summary: loss_reg: 0.355,w_loss_reg: 0.355,loss_reg_label: 0.001,w_loss_reg_label: 0.002,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.357,time_duration (s): 19.118

* Epoch: 1
Epoch summary: loss_reg: 0.346,w_loss_reg: 0.346,loss_reg_label: 0.001,w_loss_reg_label: 0.003,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.002,w_loss_regularizer: 0.002,loss: 0.35,time_duration (s): 16.981

* Epoch: 2
Epoch summary: loss_reg: 0.339,w_loss_reg: 0.339,loss_reg_label: 0.002,w_loss_reg_label: 0.004,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.344,time_duration (s): 17.122

* Epoch: 3
Epoch summary: loss_reg: 0.312,w_loss_reg: 0.312,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.001,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.323,time_duration (s): 17.084

* Epoch: 4
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.001,w_loss_regularizer: 0.001,loss: 0.32,time_duration (s): 16.975

* Epoch: 5
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.009,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 16.951

* Epoch: 6
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 18.685

* Epoch: 7
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 18.363

* Epoch: 8
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 17.373

* Epoch: 9
Epoch summary: loss_reg: 0.309,w_loss_reg: 0.309,loss_reg_label: 0.005,w_loss_reg_label: 0.01,loss_reg_lr: 0.0,w_loss_reg_lr: 0.0,loss_regularizer: 0.0,w_loss_regularizer: 0.0,loss: 0.319,time_duration (s): 17.415

Training finished #
#####

Final loss: 0.31907401164448074
Done in 523.14s.

Failed subjects: 0 / 1
(none)

-----#
All DONE #
-----#

Warnings / errors:
"clear" no se reconoce como un comando interno o externo,
programa o archivo por lotes ejecutable.
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\processing.py:76: FutureWarning: `binary\_opening` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.opening` instead.
mask = binary_opening(mask, ball(3))
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\processing.py:340: FutureWarning: `binary\_dilation` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.dilation` instead. Note the lack of mirroring for non-symmetric footprints (see docstring notes).
mask_dilated = binary_dilation(mask_symm, np.ones((10, 10, 10)))
C:\Users\Usuario\Desktop\Practicas_DaTSCAN\darq\darq\src\preprocessing.py:63: FutureWarning: `binary\_opening` is deprecated since version 0.26 and will be removed in version 0.28. Use `skimage.morphology.opening` instead.
image = binary_opening(image, self.struct_fn(data_dict[k]))